

Both star-body valuation questions in arXiv:1611.03345 are answered by arXiv:1709.08959

Literature-status note

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Status. Explicit affirmative later-paper answers to Questions 7.5 and 7.6. The later results are stated on the larger class of bounded Borel star sets and therefore imply the requested conclusions for star bodies. This note records a literature identification, not a new theorem.

1 The two source questions

Let $\mathcal{S}_0^n$ be the star bodies in $\mathbb{R}^n$, identified with their continuous radial functions on S^{n-1} , and let $\mathcal{S}_b^n$ denote bounded Borel star sets. Tradacete and Villanueva prove in arXiv:1611.03345v1 that a radial continuous valuation extends from $\mathcal{S}_0^n$ to $\mathcal{S}_b^n$ and has an integral representation on simple radial functions [1]. They then ask on PDF page 23:

Question 7.5. Is every radial continuous valuation on the star bodies of S^{n-1} uniformly continuous on bounded sets?

Question 7.6. Is the integral representation in Theorem 1.2 valid for every star body?

The source notes that an affirmative answer to Question 7.5 would imply one to Question 7.6.

2 The explicit later answers

The same authors' follow-up arXiv:1709.08959v1 is titled *Continuity and representation of valuations on star bodies* [2]. Its abstract already announces both conclusions.

First, Theorem 3.8 on PDF page 11 states that every radial continuous valuation $V : \mathcal{S}_b^n \rightarrow \mathbb{R}$ is uniformly continuous on bounded sets: for every $\lambda, \varepsilon > 0$ there is $\delta > 0$ such that

$$\|f\|_\infty, \|g\|_\infty \leq \lambda, \quad \|f - g\|_\infty \leq \delta \quad \implies \quad |\tilde{V}(f) - \tilde{V}(g)| \leq \varepsilon.$$

Restricting this theorem to continuous radial functions gives the affirmative answer to Question 7.5.

Second, Theorem 1.1 on PDF page 2 characterizes all radial continuous valuations. It says that such a valuation is exactly one for which there are a finite Borel measure μ on S^{n-1} and a strong-Carathéodory kernel $K : \mathbb{R}_+ \times S^{n-1} \rightarrow \mathbb{R}$, locally dominated by an $L^1(\mu)$ function, such that

$$V(L) = \int_{S^{n-1}} K(\rho_L(t), t) d\mu(t)$$

for every star set L with bounded Borel radial function. In particular this holds for every star body, affirmatively answering Question 7.6 in a stronger form.

Thus the exact disposition is

Questions 7.5 and 7.6: both affirmative by arXiv:1709.08959, Theorems 3.8 and 1.1.

References

- [1] P. Tradacete and I. Villanueva, *Radial continuous valuations on star bodies and star sets*, arXiv:1611.03345v1 (2016).
- [2] P. Tradacete and I. Villanueva, *Continuity and representation of valuations on star bodies*, arXiv:1709.08959v1 (2017).